# ESP-FC Next

## Dual-chip ESP32-P4 + ESP32-C5 flight controller and firmware for indoor autonomous swarm research

| Item | Value |
|---|---|
| Document | Project plan, gap analysis and Phase 1 task list |
| Version / date | v1.5 - October 27, 2026 |
| Base project | Espressif ESP-Drone (GPL-3.0), master after PR #137 (ESP-IDF 5.5 / 6.1) |
| First target | Indoor swarm of 4 drones, USA |
| Guiding rule | Nothing is ordered or assembled until math, simulation and code/PCB review gates pass |

Prices and timelines in this document are rough estimates for planning only and should be re-checked with suppliers before purchasing.

# 1. Executive summary

The goal is a cheap but highly accurate flight controller (FC) and firmware built on Espressif chips, validated in simulation before any hardware is assembled. The first application is an indoor swarm of four 2-inch ducted drones for autonomy research; a surveillance variant (camera, video streaming, GPS) follows on the same board.

- **Two chips:** ESP32-P4 runs flight control, state estimation and (later) vision; an FCC-certified ESP32-C5 module handles all radio work (Wi-Fi 6 dual-band, ESP-NOW swarm links).
- **Accurate, low-cost sensors:** ICM-42688-P IMU on SPI at up to 8 kHz, BMP581 barometer, optical flow + time-of-flight, INA226 power monitoring, bidirectional DShot RPM telemetry.
- **Shared indoor positioning:** low-cost motion capture (PS3 Eye cameras + IR markers) gives all four drones one coordinate frame and ground-truth for measuring accuracy; UWB later.
- **Simulation-first:** six verification gates (G1-G6) covering math models, software-in-the-loop, static analysis/fuzzing/AI review, PCB checks, hardware-in-the-loop, then real flight.
- **Compatibility:** keeps the Crazyflie CRTP protocol so Crazyswarm2 (ROS 2) and its simulator can be used directly for swarm research.

## Confirmed decisions

| Topic | Decision |
|---|---|
| Use case | Indoor swarm research first (4 drones); surveillance variant later |
| Location / rules | USA - indoor flight first; FCC-certified radio module |
| Scope | New FC PCB + firmware, both open (GPL-3.0 accepted) |
| Chips | ESP32-P4 (flight + vision) + ESP32-C5 module (radio) |
| Frame | 2-inch (85-90 mm) ducted brushless whoop, off-the-shelf first, custom 3D-printed later |
| Motors / ESC | 1103-1104 brushless, separate 4-in-1 ESC with Bluejay (bidirectional DShot) |
| Positioning | Low-cost mocap first + onboard flow/ToF; UWB later |
| Simulator | Own validated 6-DOF model + in-process SITL first; pybullet for obstacles/cameras; Gazebo + CrazySim / Crazyswarm2 (ROS 2) for swarm work |
| Ground control | Laptop + gamepad over Wi-Fi to start; ESP-NOW USB swarm dongle; ELRS safety radio for every real flight |
| P4 <-> C5 link | Custom SPI protocol (recommended over ESP-Hosted for ESP-NOW control and timing) |

## Prior art to benchmark

rtlopez/esp-fc (Betaflight-configurator-compatible ESP32 firmware), ArduPilot's experimental ESP32 port, Crazyflie upstream, and the open-source Low-Cost-Mocap project (ESP32 drone swarm with PS3 Eye cameras).

## Progress so far (v1.6)

| Task | Status | Key result |
|---|---|---|
| 1.1 Repo + toolchain | Done | Private GitHub repo, Docker ESP-IDF 5.5 (Colima), CI builds P4 + S3 + C5 firmware |
| 1.2 Portable flight core | Done | 18 Crazyflie algorithm files build on the PC unchanged (fc_core); HAL = existing driver interfaces |
| 1.3 Unit tests | Done | 7 Unity suites, 45 tests under ASan/UBSan; found and fixed 2 real bugs: Kalman 0/0 NaN when gyro reads exactly zero (could reset the FC in flight) and a 64-bit invSqrt over-read |
| 1.4 Gate G1 math model | Done - PASS | Stock ESP-Drone timing (500 Hz) + stock gains: roll/pitch rate loop PM 24.6 deg, GM 4.5 dB (fails). Planned 4 kHz loop: 3.5x worst-case bandwidth (11.3 vs 3.3 Hz) at equal margins; robust gains pass 1000-sample Monte Carlo (+/-20 % mass, thrust, inertia); interactive 3D replay viewer |
| 1.5 SITL | Done - G2 hover PASS | Real C flight code (fc_step + synchronous Kalman) flies the 6-DOF model in 1 ms ctypes lockstep; 84 params / 103 logs by cfclient names; steady hover error < 3 cm, estimator error < 1 cm. Findings: posCtlPid.zKi causes 11.4 cm takeoff overshoot (0.2 cm with zKi = 0); no velocity feedforward in the 2021 position controller (0.3-0.4 s tracking lag); stock link-loss behaviour drops the drone. Gym-style RL env. |
| 1.6a Flight-core simulation | Simulation checks pass; hardware pending | 1/4 kHz host variants, 500 Hz EKF at 4 kHz, G1-generated board profiles, velocity/acceleration feedforward, exact sqrt on FPU builds, arming/tumble guards. Camera-tracked figure-8: about 4-5 cm maximum error in nominal simulation (10 cm target); flow-only absolute positioning FAILS (FLOW-01). |
| 1.6b Motor vibration + RPM notch | Simulation checks pass; hardware pending | Synthetic motor-vibration model (ASSUMED amplitudes); 4-motor 1x/2x RBJ notch bank, 100 Hz RPM telemetry, 100 ms bypass. 12 deg/s assumed artifact: 3.8 -> 0.20 deg/s RMS; motor activity >250 Hz 3.1% -> 1.9% (baseline 1.9%). Hover/figure-8 tracking unchanged (LPF already attenuates the band). 500 Hz notch floor; worst sampled RPM/32-corner margin 46.2 deg / 6.9 dB (hover only: 47.5 deg / 7.2 dB). |
| Safety: hold + landing | Opt-in SITL; hardware pending | Command loss: fixed-pose hold then descent; camera loss: level descent with independent local ToF. LAND stays latched through touchdown, even if commands return; terminal states block automatic rearm. Nominal 1/4 kHz SITL: max descent 0.263 m/s, sampled touchdown approach 0.093-0.102 m/s; motors off at ground. All-height loss is a separate FAULT/motor stop, not safe landing. |
| Local flow braking | SITL checks pass; hardware pending | Healthy flow/range enables zero-velocity braking after camera loss instead of passive coasting. Nominal excursion 1.71-1.72 m -> 0.277-0.290 m; sampled stress worst 0.349 m at 1/4 kHz. Stale flow degrades to level descent; restored jumped camera frames are not reacquired. No absolute-positioning or total-height-loss solution claimed. |
| Input plausibility | Done (host/SITL) | Audit found one NaN IMU sample poisoned the EKF permanently and tripped asserts (motor stop on target) in both modes. Bad IMU samples are now held for up to 5 ms, then disarm/FAULT; non-finite position/ToF/flow are dropped in legacy mode too. Regression tests failed before the fix. |
| 1.7 Link protocols | In progress | docs/protocols.md: 64-byte P4-C5 SPI frame with CRC-16, 20-byte 50 Hz swarm status, 32-message anti-replay window; unit tests + libFuzzer (make link-fuzz); builds on the C5 toolchain. AES-CCM envelope specified, not yet built/tested; no SPI/ESP-NOW transport yet. |
| 1.9 G3 quality gate | Started | cppcheck clean (make core-lint, CI); fc_core sources 91-100% line coverage, but the compiled Crazyflie set is 55% overall (unused INDI/Mellinger/planner at 0-5%); clang-tidy not set up. |

4 kHz is a host/SITL configuration, not a measured ESP32-P4 runtime. The production P4 image still uses the existing ESP-Drone task loop. Driver integration, CPU deadlines, Wi-Fi/camera contention and hardware flight are unvalidated. Dynamic-notch/RPM filtering was added in Task 1.6b as a simulation-only model.

Task 1.6b is a synthetic stress model, not measured vibration. It adds a gyro artifact only, not frame resonance or body torque. The notch reduces high-frequency gyro and motor-command activity but did not improve position tracking in simulation; it does not fix FLOW-01 (flow-only absolute XY drift). Review found the initial 150 Hz notch floor failed off-hover margins; it is now 500 Hz per section. Q=5 sampled linear checks are not an all-speed or hardware guarantee. RPM calibration, pole-pair conversion, ESC telemetry latency and real vibration spectra require hardware blackbox data. The 4 kHz runtime remains host/SITL only; the production P4 image still uses the existing ESP-Drone task loop.

Latched landing is opt-in in the portable runtime, not activated on hardware. Thresholds and touchdown dwell are engineering assumptions. Without local flow, moving camera-loss tests drift about 1.7-2.0 m. Flow/range braking reduces the tested excursion but does not provide collision-safe swarm flight; all-height loss can still cause a fall. Finite frozen/replayed sensor data is not detected. The four-drone fault/landing acceptance gate remains open; nominal single-drone tests do not pass it. BMP581 pressure noise is not equivalent to altitude accuracy: barometer/contact redundancy is still research, not an implemented safe-touchdown fallback.

# 2. ESP-Drone gap analysis

Reviewed: the ESP-Drone documentation (Get Started, Hardware Reference, Drivers, Flight Control System, Communications) and the GitHub repository. Note: upstream merged a large refactor on Oct 26, 2026 (PR #137) adding ESP-IDF 5.5/6.1, ESP32-P4/S31 targets, disarmed-by-default startup, NVS configuration storage and malformed-packet rejection. Items partly fixed by it are marked *(partly fixed upstream)*.

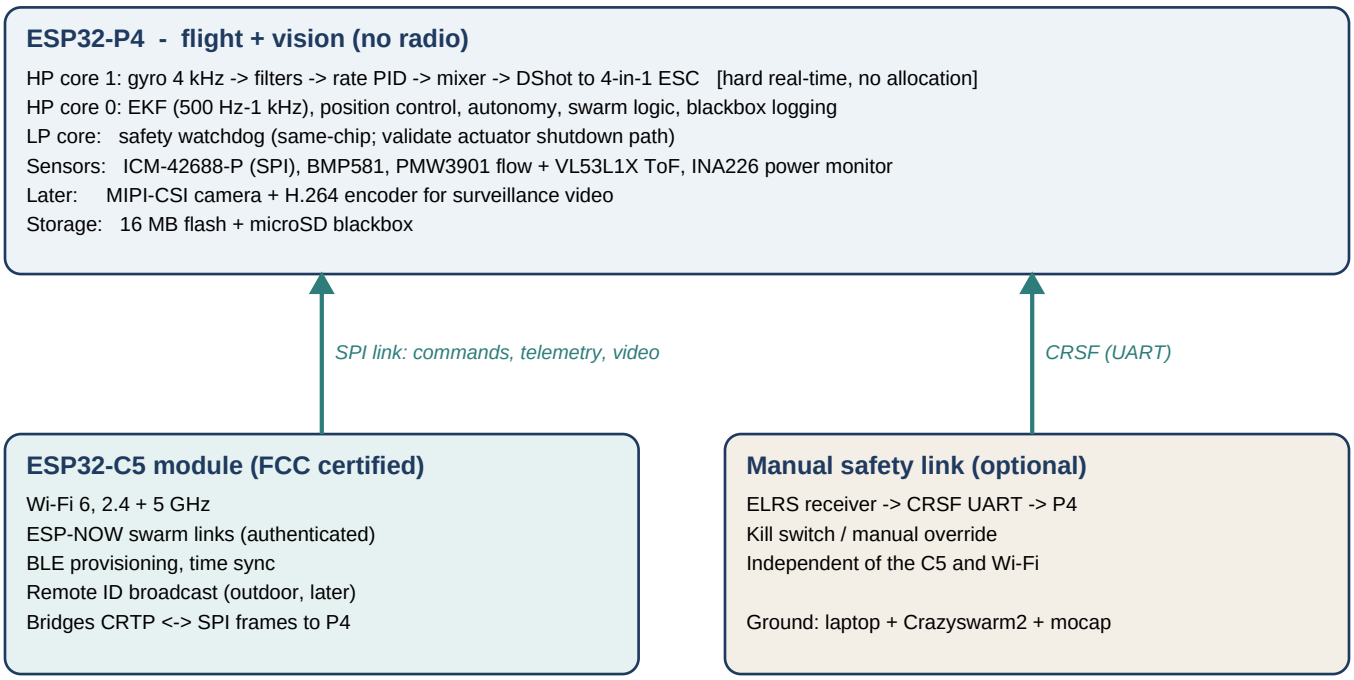
## 2.1 Hardware

| Area | Current state | Gap | Opportunity |
|---|---|---|---|
| Main chip | ESP32-S2 (single core) on reference board | Kalman estimator must run at low priority or the watchdog fires | Dual-chip P4 + C5: dedicated real-time core, radio isolated |
| IMU | MPU6050 on I2C, as low as 125 Hz sampling | Old, noisy, slow bus, single unit | ICM-42688-P on SPI, 4-8 kHz, data-ready interrupt, FIFO; optional 2nd IMU |
| Baro / compass | MS5611, HMC5883 (discontinued) | Baro altitude hold reported broken in open issues | BMP581 barometer; LIS3MDL compass (optional) |
| Motors | Brushed 716/720 only, simple PWM | No brushless/ESC support, no RPM telemetry | DShot300/600 via RMT, bidirectional DShot RPM |
| Power | 1S, voltage via VBAT/2 divider | No current sensing; brownout issues reported | INA226, separate logic/motor rails, bulk caps, reverse-polarity protection |
| Logging | None onboard | No post-flight vibration/tuning analysis | Blackbox to flash / microSD |
| Positioning | Flow + ToF, indoor only | No GPS (issue open since 2022); no shared swarm frame | Mocap + flow; UWB; M10 GPS later |
| RC input | Wi-Fi app / gamepad only | No standard RC receivers | CRSF / ELRS over UART |
| Openness | Non-KiCad design files | KiCad export requests; S3/V2 schematics missing | Fully open KiCad design |

## 2.2 Firmware

| Area | Gap | Opportunity |
|---|---|---|
| Codebase age | Flight core is Crazyflie 2021.01 (about 5 years old) | Rebase on newer Crazyflie features: arming supervisor, controllers, Kalman fixes |
| Loop timing | I2C IMU; flight loop competes with Wi-Fi tasks | Flight loop pinned to its own core at 2-4 kHz with measured jitter |
| Filtering | Only a basic 2-pole low-pass | Dynamic notch (FFT), RPM notch, D-term LPF, feedforward, airmode |
| Control | Only PID tuned; Mellinger/INDI untested | Tuned INDI; acro/rate modes |
| Failsafe | Basic link-timeout cutoff (disarmed by default: partly fixed upstream) | Auto-land on link loss / low battery, crash detection, geofence, preflight checks |
| Security | Fixed AP password 12345678; no command authentication; 1-byte checksum; no replay protection | Per-device credentials, WPA3, AES-CCM authenticated frames with sequence numbers |
| Radio links | Phone over Wi-Fi with variable latency | ESP-NOW low-latency swarm links; 5 GHz on the C5 |
| Protocols | CRTP only; forked cfclient hangs (open issues) | Keep CRTP for cflib/Crazyswarm2; add MAVLink and micro-ROS later |
| Updates | No OTA | ESP-IDF OTA, secure boot, flash encryption |
| Testing | No SIL / HIL / unit tests | Host build, simulator bridge, fuzzing, HIL in CI |
| Docs / support | Some 404 pages; limited support since Dec 2022 | Maintained docs as part of this project |

# 3. System architecture




## Why this split

- **ESP32-P4:** large compute headroom over the S2/S3, a low-power safety monitor (same-chip; validate actuator shutdown path), and a camera interface plus hardware H.264 encoder for the surveillance variant. The LP core is NOT an independent safety barrier.
- **ESP32-C5 module:** dual-band Wi-Fi 6 (5 GHz is less crowded indoors) and ESP-NOW for drone-to-drone messages. Radio interrupts never touch the flight loop. A pre-certified module can reduce FCC certification work only within its grant and integration constraints; host compliance obligations remain.
- **Fills an upstream gap:** ESP-Drone currently runs the P4 in a no-radio diagnostic profile and notes it still needs a companion radio design.
- **Failure isolation:** if the C5 or Wi-Fi fails, the P4 keeps stabilizing and runs its failsafe (hover, then land); the ELRS link remains available as a manual override.

## Firmware structure

| Layer | Contents | Runs on |
|---|---|---|
| fc_core (portable C library) | Filters, estimator (EKF), rate/attitude/position control, mixer, commander, safety supervisor. No FreeRTOS calls, no dynamic allocation. | P4 and PC (SITL / unit tests) |
| HAL interface | Time, sensors, motors, link, storage. One implementation per target (P4 board, simulator). | P4, PC |
| Drivers | ICM-42688-P, BMP581, PMW3901, VL53L1X, INA226, DShot (RMT), CRSF, SPI link | P4 |
| Link firmware | SPI frame bridge, ESP-NOW swarm, Wi-Fi UDP (CRTP), time sync, provisioning | C5 |
| Tools | Swarm dongle firmware, ground scripts, Crazyswarm2 config, web configurator (later) | C5 dongle, PC |

# 4. Indoor swarm drone (Rev A build)

## 4.1 Airframe

| Option | Verdict |
|---|---|
| Brushed micro (ESP-Drone / Crazyflie style) | Cheap, but brushed motors wear unevenly so the four drones drift apart in behavior; little payload. |
| 2-inch ducted brushless whoop (SELECTED) | Ducts protect props and people in frequent swarm crashes; cheap spares in US stores (GetFPV, RaceDayQuads, BetaFPV, NewBeeDrone); payload for flow, ToF and markers. |
| 3-inch+ open props | Too dangerous and loud for indoor swarm work. |

| Part | Selection | Notes |
|---|---|---|
| Frame | 2-inch (85-90 mm) ducted, 20x20 mm FC mount (or printed adapter) | Custom 3D-printed TPU frame later |
| Motors | 1103-1104 brushless | 40 mm props |
| ESC | Off-the-shelf 4-in-1, Bluejay firmware | Bidirectional DShot; replaceable separately |
| Battery | 1S HV 450-550 mAh (2S optional) | Choose after weight check in simulation |
| Weight target | 50-70 g all-up | Well under 250 g |
| Markers | 3-4 IR reflective markers, unique pattern per drone | For mocap identification |

## 4.2 Flight controller sensors and parts

| Function | Part | Reason |
|---|---|---|
| IMU | ICM-42688-P (SPI) | Low noise, 8 kHz, low cost; footprint for a second IMU |
| Barometer | BMP581 | Low noise altitude reference |
| Flow / height | PMW3901 (or PAA3905) + VL53L1X | Indoor hover; drivers exist in ESP-Drone |
| Power monitor | INA226 + shunt | Accurate battery state for failsafes |
| Storage | 16 MB flash + microSD | Blackbox logging |
| Radio | ESP32-C5 certified module | Wi-Fi 6 dual-band, ESP-NOW |
| Optional | UWB (DW3000) footprint, LIS3MDL compass, MIPI-CSI connector | Future swarm self-localization and surveillance |

## 4.3 Indoor positioning

| Layer | What | Approx. cost | Role |
|---|---|---|---|
| Onboard | Optical flow + ToF | ~$15 / drone | Smooth local hover; backup during mocap dropouts |
| External (first) | Low-cost mocap: 4x PS3 Eye cameras + IR markers | ~$60-100 total | One shared coordinate frame for all drones; ground truth for accuracy measurement |
| Later | UWB tags + anchors (DW3000) | ~$25 / drone + anchors | Decentralized self-localization without cameras |

Data path: mocap -> ROS 2 -> Crazyswarm2 -> ESP-NOW swarm dongle -> drones. Onboard, the EKF fuses mocap position, flow, ToF and IMU so a drone keeps flying through short tracking losses.

# 5. Control and communication

| Stage | Control method | Approx. cost | Why |
|---|---|---|---|
| Simulation and bench | Laptop + gamepad over Wi-Fi | $0 | Easiest; works with existing Crazyflie tools |
| Swarm | Laptop + USB ESP32-C5 swarm dongle (ESP-NOW) | ~$8 | One radio reaches all drones with low latency |
| Every real flight | ELRS handheld radio as kill switch / manual override | ~$40-60 | Works even if Wi-Fi or the C5 fails |
| Demos | Phone app | $0 | Nice to have, not primary |

## Link protocol requirements

- P4 <-> C5: framed SPI protocol with length, type, sequence number and CRC-16; separate priority lanes for commands, telemetry and bulk data (video later).
- Air links: CRTP carried over ESP-NOW and UDP; every command frame authenticated (AES-CCM) with a monotonically increasing sequence number to block replay.
- Swarm status broadcast at 50 Hz: ID, position, velocity, battery, flight state, timestamp.
- Time sync from the dongle so all drones share a common clock for coordinated maneuvers.
- Failsafe timing: no valid command for 0.5 s -> hold position; 2 s -> controlled landing (values to be tuned in simulation).

# 6. Verification gates (test before you build)

No purchase or assembly happens until the previous gate passes. Pass criteria are initial targets and may be refined after the G1 model is built.

| Gate | What is verified | Tools | Pass criteria (proposed) |
|---|---|---|---|
| G1 Math | Rigid-body model, control loop margins, sensor noise impact, estimator behavior | Python: numpy, scipy, python-control | Phase margin > 45 deg; stable over +/-20% mass and thrust variation |
| G2 SIL | Real flight C code compiled for PC, flying in simulation; 4-drone swarm | gym-pybullet-drones, Gazebo + CrazySim, Crazyswarm2 | Hover within +/-5 cm; figure-8 error < 10 cm; 4 drones with zero collisions |
| G3 Code | Warnings as errors, static analysis, unit tests, sanitizers, parser fuzzing, AI code review | clang-tidy, cppcheck, CTest, ASan/UBSan, libFuzzer | Zero warnings; > 80% coverage of control code; 1 h fuzzing without crashes |
| G4 PCB | ERC/DRC, scripted netlist checks (strapping pins, pull-ups, power budget), power-stage simulation, DFM, AI schematic review | KiCad, ngspice, Python | No brownout in simulation at full motor current spikes; zero DRC errors |
| G5 HIL | Real P4 running firmware with simulated sensor data; motors disconnected | USB bridge to simulator | Flight-loop jitter < 5 us; 4 kHz loop holds with Wi-Fi and logging active |
| G6 Flight | Props off -> tethered -> single hover -> 2 drones -> 4 drones | Blackbox logs + mocap ground truth | Real logs match simulation within agreed tolerance |

# 7. Roadmap

| Phase | Scope | Gate(s) | Approx. cost | Approx. time |
|---|---|---|---|---|
| 1 - Software foundation | Repo, portable flight core, PC build, math model, SITL, 4-drone swarm simulation, protocol specs | G1-G3 | $0 | 4-6 weeks |
| 2 - Dev-board bring-up | P4 dev board + sensor breakouts; drivers, P4<->C5/C6 link, safety watchdog, HIL | G5 | ~$120-180 | 3-5 weeks |
| 3 - Custom FC board | KiCad P4 + C5 design, review, order 5 assembled boards (4 drones + spare) | G4 | ~$25-40 / board (small volume) | 4-6 weeks |
| 4 - Swarm flight | Build 4 drones, mocap setup, ESP-NOW swarm, Crazyswarm2 missions, optional UWB | G6 | Frames, motors, ESCs, batteries, mocap | 4-8 weeks |
| 5 - Surveillance variant | Camera, H.264 over 5 GHz, GPS, outdoor failsafes, Remote ID | G1-G6 repeated | TBD | TBD |

# 8. Phase 1 task list (software only)

Goal: a 4-drone swarm flying in simulation on the real flight code, passing gates G1-G3, before any hardware is bought.

| # | Task | Output | Done when |
|---|---|---|---|
| 1.1 | Repo + toolchain: monorepo (firmware/ sim/ hardware/ tools/ ci/ docs/), fork esp-drone master (post-PR #137), Docker with pinned ESP-IDF 5.5, GitHub Actions building P4 + C5 firmware | Working repo + CI | CI passes on every push |
| 1.2 | Separate flight code from hardware: move estimator, controller, mixer, filters, commander and supervisor into fc_core behind a HAL | fc_core library | Builds for ESP32 and PC |
| 1.3 | PC build + tests: CMake host target, unit test framework, ASan/UBSan, coverage | Test harness | Tests run in CI |
| 1.4 | G1 math model: 6-DOF dynamics with frame mass/inertia, motor lag, thrust curve; datasheet noise models; stability margins; Monte Carlo | sim/model + report | Phase margin > 45 deg; robust to +/-20% |
| 1.5 | G2 SIL: synchronous fc_step() loop (sensor calibration + LPF -> Kalman -> commander watchdog -> controller -> mixer) shared by SITL and the P4; in-process ctypes lockstep with the validated 6-DOF model (one library copy per drone for swarms); firmware PARAM/LOG access by name; Gym-style environment for RL; pybullet kept for obstacles/camera rendering later | SITL bridge + RL env | Single drone hovers within +/-5 cm |
| 1.6 | Flight-core upgrades: 4 kHz rate loop, 500 Hz-1 kHz estimator, dynamic notch + RPM hook, newer arming supervisor, mocap + flow EKF fusion | Updated core | Figure-8 error < 10 cm in simulation |
| 1.7 | Link protocols: P4<->C5 SPI framing, CRTP over ESP-NOW, AES-CCM + sequence numbers, 50 Hz swarm broadcast | docs/protocols.md + codec library | 1 h fuzzing without crashes |
| 1.8 | 4-drone swarm simulation with Crazyswarm2: takeoff, formation, position swap, landing; inject link loss, low battery, mocap dropout | Swarm sim scripts | All missions complete; every fault ends in a safe landing |
| 1.9 | G3 quality gate: clang-tidy, cppcheck, warnings as errors, AI review checklist, timing budget | CI report | Zero warnings; > 80% control-code coverage |

# 9. AI and autonomy track

Learned components output setpoints (position/velocity/attitude targets), never raw motor commands; the verified PID loop and safety supervisor remain the fallback. Training runs on a workstation, with simulation validation under the same gates. On-board inference only after validation.

## What fits on the ESP32-P4

| Task | Model | Reported P4 performance | Fit |
|---|---|---|---|
| Person / object detection (surveillance) | ESPDet-Pico, 224x224 input (Espressif esp-detection) | About 51 ms inference, over 18 FPS incl. pre/post-processing | Good: detect-and-follow at low speed |
| General multi-class detection | YOLO11n, 640x640 | About 2.75 s per frame | Too slow for flight; use for offline/ground processing |
| RL flight/swarm policy | Small MLP (e.g. 2 x 64 neurons) | Not benchmarked; depends on network size, quantization and concurrent load | Candidate for high-level setpoints; benchmark before deployment |

Detection figures are published Espressif / community benchmarks, not yet measured on this hardware; the flight loop occupies one P4 core, so CNN frame rate will be measured in Phase 2.

## Roadmap

| Phase | AI work |
|---|---|
| Phase 1 (now) | SITL exposes a Gym-style environment (reset/step) so policies can be trained on a workstation |
| Phase 4 (swarm) | Workstation-trained RL for formation keeping and collision avoidance in the 4-drone simulation. On-board MLP inference from ESP-NOW neighbour states only after validation and benchmarking |
| Phase 5 (surveillance) | On-board person detection on the P4 camera input (MIPI-CSI) with ESP-DL; detect-and-follow behaviour gated by the safety supervisor |

## Safety rules for learned components

- Outputs are bounded setpoints only.
- The supervisor can always override (geofence, attitude/rate limits, battery).
- Every policy must pass the same SIL/HIL gates as hand-written code.
- Policy inputs and outputs are logged in the blackbox.

# 10. Phase 2 shopping list (buy only after G3 passes)

| Item | Qty | Purpose |
|---|---|---|
| ESP32-P4-Function-EV-Board (includes ESP32-C6) | 1 | Two-chip development before the custom board |
| ICM-42688-P breakout | 1 | IMU driver and timing tests |
| BMP581 breakout (e.g., SparkFun) | 1 | Barometer driver |
| PMW3901 + VL53L1X breakouts | 1 each | Flow and height drivers |
| INA226 breakout | 1 | Power monitor driver |
| ESP32-C5 DevKit | 1 | Swarm USB dongle firmware |
| PS3 Eye cameras | 2 (4 later) | Early mocap testing |
| Logic analyzer (8 ch, low cost) | 1 | Measure loop timing, SPI and DShot signals |

Estimated total: about $120-180 before the custom board.

# 11. US regulatory notes

| Topic | Notes |
|---|---|
| Indoor flight | FAA airspace rules generally do not apply to flights fully indoors; no registration or Remote ID needed for indoor-only research. |
| Radio (FCC) | Use an FCC-certified ESP32-C5 module (not the bare chip) within its grant and integration constraints (antenna, placement, RF exposure and labeling). Module certification is not an exemption from all FCC host testing; host emissions, RF exposure and other applicable compliance obligations remain. The P4 has no radio. Verify certification and integration requirements for the selected ELRS gear. |
| Outdoor (later) | Under 250 g recreational: registration not required unless flying under Part 107. Over 250 g or research under Part 107: registration and Remote ID required (the C5 can broadcast it). |
| Surveillance | State privacy and recording laws apply. Fly only in spaces you control during research. |

This is general guidance, not legal advice; verify current FAA/FCC rules before outdoor flight.

# 12. Risks and mitigations

| Risk | Impact | Mitigation |
|---|---|---|
| ESP32-P4 silicon revision / ESP-IDF maturity | Driver gaps or errata | Bring up on the official dev board first; pin ESP-IDF version; track errata |
| P4 <-> C5 link complexity | Schedule slip | Spec and fuzz the protocol in Phase 1; fallback to ESP-Hosted for Wi-Fi-only operation |
| Weight budget on a 2-inch frame | Short flight time | Weight model in G1; 2S option; minimize optional parts on swarm build |
| Mocap occlusion / reflections | Position dropouts | EKF fusion with flow + ToF; hold-then-land failsafe; matte room surfaces |
| 2.4 GHz interference indoors | Link latency | Use 5 GHz on the C5; ESP-NOW channel planning; ELRS for safety link |
| Higher power draw of a two-chip design | Battery life | Measure in Phase 2; power down unused peripherals |
| GPL-3.0 obligations | Must publish source of distributed firmware | Accepted; keep all changes in a public repo |
| Crashes during swarm testing | Hardware damage, injury | Ducted frames, ELRS kill switch, netted test area, simulation first |
| Learned policies behaving unexpectedly outside training conditions | Unsafe commands | Setpoint-only interface, supervisor override, domain randomization in simulation, SIL/HIL gates before flight |

# 13. Immediate next steps

- Create the monorepo and fork ESP-Drone master into firmware/ (Task 1.1).
- Set up the Docker toolchain with ESP-IDF 5.5 and CI builds for ESP32-P4 and ESP32-C5.
- Start the Python 6-DOF model using measured specs of the selected 2-inch frame (Task 1.4).
- Draft docs/protocols.md for the P4 <-> C5 link and swarm messages (Task 1.7).